

MTH204 (Section A)
Graded Worksheet–7

Name:

Roll No.:

Important Notice: The use of electronic gadgets is strictly prohibited during the tutorial. This includes mobile phones, tablets, or any device containing class notes, but anything handwritten is completely allowed.

Q-1) Solve the ODE

$$y'' - y' - x^2y = 0$$

using the power series method.

Q-2) Find a solution of

$$(a^2 - x^2)y'' - 2xy' + n(n+1)y = 0$$

by reduction to a Legendre equation.

Q-3) Solve

$$xy'' + y' + \frac{1}{4}y = 0$$

by making the change of variable $z = \sqrt{x}$.

Q-4)

(a) Evaluate the integral:

$$\int x^{-1} J_4(x) dx.$$

(b) Prove that:

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x.$$

Q-5) Find a basis of solutions using the Frobenius method for the differential equation:

$$2xy'' + (3 - 4x)y' + (2x - 3)y = 0$$

Solution:

Solution:

Solution:

Solution:

Solution: